Equivalent Modeling of Cloud Data Center Overlay Digital Twins Based on Instance Deduplication

Chi Qingyuan[1], Ruslan L. Smeliansky[1]

Faculty of Computational Mathematics and Cybernetics, Lomonosov Moscow State University, Moscow 119991, Russia (E-mail: chi.q@asvk.cs.msu.ru smel@cs.msu.su)

Abstract: With the rapid evolution of Intent-Driven Networks (IBN) powered by Large Language Models (LLMs) in cloud data centers (DCs), digital twins have emerged as a critical sandbox to verify network intents and prevent LLM configuration hallucinations. However, traditional 1:1 full-scale simulation based on Underlay physical topologies suffers from severe resource explosion and verification latency. This is primarily due to the massive multiplication of physical instances introduced by distributed anycast gateways (EVPN-VXLAN). To achieve second-level closed-loop verification, this paper proposes a lightweight digital twin modeling method oriented toward the Overlay logical layer.

Capitalizing on the transparency of VXLAN protocols to upper-layer services, this approach strips away underlying physical details and performs "logical deduplication" on Virtual Routing and Forwarding (VRF) and Bridge Domain (BD) instances scatteredly deployed across multiple physical Leaf switches. We construct a mathematical model for simulation overhead under equivalent service verification constraints, theoretically demonstrating that the state space complexity is reduced from $\mathcal{O}(N^2 + T \cdot N)$ to $\mathcal{O}(T)$, where N denotes the physical network scale and T denotes the total tenant scale.

Multi-dimensional Monte Carlo simulations show that while ensuring complete equivalence in connectivity and isolation verification, the proposed method achieves up to a 61.78-fold resource space compression in a large-scale network scenario featuring 100 Leaf switches and 300 tenants. Consequently, this renders the twin state space fully immune to physical node expansion with an O(1) constant property. This lightweight infrastructure significantly improves the verification efficiency of digital twins, providing an ultra-fast and low-overhead anti-hallucination verification foundation for future LLM-driven IBN toward second-level autonomous closed-loop operation.

Key words: Digital Twin; Intent-Driven Network (IBN); Cloud Data Center; Overlay Abstraction; Logical Instance Deduplication

1. Introduction

In recent years, the deep integration of Intent-Driven Networking (IBN) and Large Language Models (LLMs) has emerged as a core paradigm shift for next-generation intelligent network operations (NetOps)[1]. Leveraging their powerful natural language understanding capabilities, LLMs can directly translate abstract service intents into concrete network configuration scripts. However, generative large models suffer from inherent "hallucinations," making their generated configuration instructions highly susceptible to high-risk anomalies such as routing conflicts and isolation failures. Therefore, before network configurations are officially deployed to the production network, pre-compliance verification within a digital twin simulation environment is strictly required.

In modern cloud data center scenarios, network architectures universally adopt a Spine-Leaf topology based on distributed anycast gateways, such as EVPN-VXLAN[2]. To support the large Layer 2 roaming of virtual machines (VMs) across arbitrary physical servers, instances of the same tenant's

Virtual Private Cloud (VPC), Virtual Routing and Forwarding (VRF) domains, and Layer 2 Bridge Domains (BDs) are highly scattered and instantiated across multiple physical Leaf switches. In traditional 1:1 Underlay full-scale physical digital twin modeling, to verify configuration correctness, the system must launch corresponding network operating system (NOS) images or processes for every single physical device. Furthermore, it must fully simulate the full-mesh VXLAN tunnels across the underlying network[3].

This traditional full-scale simulation suffers from a severe "physical replica explosion" problem. For instance, consider a typical tenant (VPC) in a cloud data center designed with 1 logical VRF and 2 logical BD subnets. Suppose BD1 connects 4 VMs and BD2 connects 3 VMs. Due to the randomness of cloud computing scheduling, these 7 VMs happen to be scattered across 7 different physical Leaf switches. If a 1:1 physical full-scale simulation is adopted, the system must run identical VRF processes, BD instances, and IRB Layer 3 interface configurations individually on 7 distinct virtual switch instances. When the entire network contains T tenants and the VMs of each tenant are highly dispersed, the number of physical configuration instances inside the twin sandbox will explode exponentially with the product of the number of tenants and devices. Consequently, constructing such a digital twin demands massive hardware computing resources, and the verification cycle is prolonged to the minute level. This thoroughly negates the practical value of using digital twins to safeguard second-level closed-loop intent execution in LLM-driven IBN.

To address this challenge, this paper proposes a lightweight digital twin modeling method oriented toward the Overlay logical layer. The core contributions of this paper are summarized as follows:

① We propose a lightweight Overlay digital twin architecture for cloud data centers based on logical instance deduplication that strips away the underlying Underlay physical dependencies. This architecture completely decouples the simulation overhead from the scale of physical switches.

② We establish a state space mathematical model under equivalent service verification constraints, rigorously proving that the proposed method reduces the complexity from $\mathcal{O}(N^2 + T \cdot N)$ to $\mathcal{O}(T)$.

③ Through multi-dimensional Monte Carlo randomized simulations, we quantitatively confirm that the proposed method achieves over a hundredfold compression of the resource state space in large-scale network environments, demonstrating complete immunity to physical topology expansion.

2. Architecture Mapping and Service Equivalence

2.1 Lightweight Overlay Mapping Architecture

The lightweight modeling principle proposed in this paper is summarized as "abstracting the underlying layer, deduplicating the higher layer, and ensuring service equivalence." This method breaks away from the traditional reverse modeling approach of "constructing physical devices before deploying service configurations." Instead, it directly builds logical twin entities based on higher-layer service entities, including VPCs, VRFs, and BDs.

The specific mapping principles are defined as follows: A tenant's Virtual Private Cloud (VPC) is abstracted into a globally unique, deduplicated logical node of a Layer 3 routing isolation domain (logical VRF). Subnets within the VPC are abstracted into deduplicated logical nodes of Layer 2 broadcast domains (logical BDs). Virtual Machines (VMs), serving as terminal nodes, are completely decoupled from their physical host tags and directly attached to their corresponding logical BD nodes.

In this lightweight Overlay twin model, physical Leaf switch entities and the full-mesh physical VXLAN tunnels interconnecting them are entirely eliminated. The mapping metadata, which tracks how VRF and BD instances are scatteredly deployed across specific physical Leaf switches in the production network, is maintained within an in-memory mapping table by the SDN controller and is not deployed as independently running entities inside the digital twin sandbox [4].

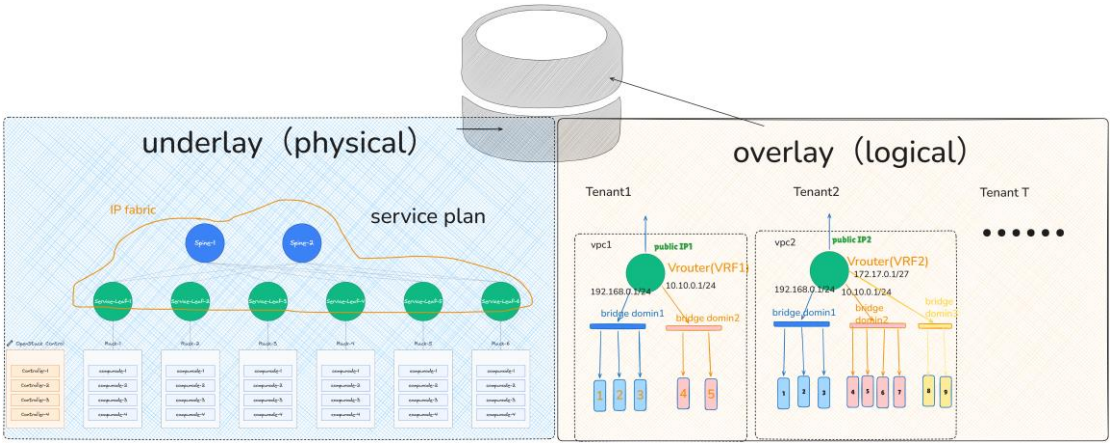


Fig.1. Architectural comparison and conceptual schematic between the 1:1 Underlay full-fidelity physical modeling and the Overlay logical deduplication modeling in cloud data centers.

2.2 Strict Definition of Service Equivalence and Justification

To guarantee that the lightweight model can accurately intercept configuration hallucinations generated by LLMs after discarding underlying physical details, we provide a mathematically rigorous definition of "service equivalence."

Let $\boxtimes$ denote the set of intent-driven network security policy verification tasks generated by the LLM (including checking inter-tenant VPC hard isolation, subnet internal connectivity, cross-subnet routing reachability, and ACL access control policy compliance). Let M_u represent the traditional full-scale physical digital twin model, with its corresponding intent decision function denoted as $f_u(r), r \in \boxtimes$. Let M_o represent the deduplicated logical digital twin model proposed in this paper, with its decision function denoted as $f_o(r)$. The two models are defined as service equivalent if and only if their decision results for any arbitrary task are completely identical:

$$\mathrm{Equiv}(M_u, M_o) \Leftrightarrow \forall r \in \boxtimes : f_u(r) = f_o(r)$$

The core underlying assumption of this model is the "neglection of Underlay link congestion and underlying encapsulation details." This assumption is firmly justified by the current operational realities of industrial cloud data centers. Modern data centers universally employ highly redundant two-tier or three-tier CLOS topologies and run Equal-Cost Multi-Path (ECMP) routing to achieve uniform load balancing across the entire network fabric[5].

Empirical measurement data from production data centers indicate that the utilization of underlying links generally remains within a low-load, safe zone; the average utilization of edge links is typically below 10%, while the median utilization of aggregation links ranges from approximately 10% to 20%[6]. Since the VXLAN encapsulation mechanism itself is entirely transparent to upper-layer service data streams, underlying routing perturbations and physical encapsulations do not alter the logical correctness of upper-layer service configurations (such as misconfigured IPs or missing ACLs).

Therefore, stripping away these underlying physical details does not cause any degradation in the detection accuracy of upper-layer service connectivity and isolation verifications[7].

3. Mathematical Model for Simulation Overhead

To provide a rigorous theoretical foundation explaining the root cause of resource preservation, this section establishes asymptotic complexity state space models for both twin paradigms. Assume the network consists of N physical Leaf (VTEP) nodes, and the current total number of tenants (VPCs) in the production network is T. For the i-th tenant（$1 \le i \le T$）, the number of planned logical BDs is denoted as b_i.

3.1 Overhead of Traditional Underlay 1:1 Full-Fidelity Model(M_u)

In full-scale physical simulation, the digital twin system must construct individual replicas for each tenant's gateway instances on every physical Leaf switch where they are scatteredly deployed in reality[8]. In the worst-case scenario where tenant applications are maximally dispersed due to live migration, let $S_{bd}(i,j)$ denote the set of physical Leaf switches where the j-th BD of the i-th tenant is actually deployed, and let $S_{vrf}(i)$ denote the set of physical Leaf switches hosting the corresponding Layer 3 VRF. The total runtime state space resource overhead S_u for the full-scale model M_u can be formulated as follows:

$$S_u = \sum_{i=1}^{T} \left(\sum_{j=1}^{b_i} |S_{bd}(i,j)| \right) + \sum_{i=1}^{T} |S_{vrf}(i)| + N \cdot C_{leaf} + \frac{N(N-1)}{2} \cdot C_{tun}$$

Where $|\cdot|$ represents the cardinality of the set (i.e., the absolute number of physical Leaf switches across which that specific service instance is scattered); C_{leaf} denotes the baseline constant overhead of running a single physical switch's native NOS operating system; $\frac{N(N-1)}{2}$ represents the total number of full-mesh VXLAN tunnels generated by the pairwise interconnection of N physical nodes; and C_{tun} represents the constant overhead required to maintain the dynamic protocol states of each physical tunnel within the twin data plane.

According to the operational characteristics of distributed anycast gateways, as the physical network scale N and the tenant scale T expand simultaneously, the physical deployment terms $|S_{bd}|$ asymptotically approach N， This causes the number of physical configuration replicas in the first two terms to explode at a composite scale of $\mathcal{O}(T \cdot N)$ Concurrently, the final term representing physical tunnel overhead exhibits a rigid quadratic growth of $\mathcal{O}(N^2)$ due to its full-mesh nature. Therefore, the upper bound of the overall state space complexity for the full-scale model M_u is expressed as:

$$S_u = \mathcal{O}(N^2 + T \cdot N)$$

3.2 Overhead of the Proposed Overlay Logical Deduplication Model(M_o)

In the Overlay twin model M_o, all physical nodes and full-mesh tunnels are completely decoupled from the underlying layer. Furthermore, global service deduplication is performed on identical configurations across different physical switches. Consequently, no matter how widely a BD or VRF is distributed across the production network, it is instantiated exactly once within the twin sandbox. Its total state space overhead S_o is defined as:

$$S_o = \sum_{i=1}^{T} (b_i + 1) \cdot C_{obj} + E_{logic} \cdot C_{link}$$

Where C_{obj} represents the unit constant overhead of a single pure logical service object (VRF or BD) on the deduplicated infrastructure; E_{logic} denotes the total number of basic topological logical links between VMs and their corresponding logical BDs within tenants; and C_{link} represents the constant maintenance overhead of a single logical link.

In actual cloud data centers, the number of subnets b_i and the number of VMs defined by a single tenant are governed solely by the tenant's own application architecture. These parameters remain finite small constants that do not scale with the physical device count N (typically bounded between 2 and 6)[9]. Consequently, the average logical overhead per tenant can be consolidated into a constant K, simplifying the equation to:

$$S_o = \sum_{i=1}^{T} K = K \cdot T$$

This theoretically proves that the space complexity of the deduplicated logical model M_o scales strictly linearly with the tenant capacity, completely decoupling from the number of physical switches N:

$$S_o = \mathcal{O}(T)$$

3.3 Derivation of Theoretical Compression Ratio

To intuitively evaluate the reduction efficiency of the proposed lightweight method on the twin state space, we define the theoretical Compression Ratio CR as the ratio of the traditional full-scale model overhead S_u to the proposed overlay logical deduplication model overhead S_o:

$$CR = \frac{S_u}{S_o} = \frac{\sum_{i=1}^{T} \left(\sum_{j=1}^{b_i} |S_{bd}(i,j)| \right) + \sum_{i=1}^{T} |S_{vrf}(i)| + N \cdot C_{leaf} + \frac{N(N-1)}{2} \cdot C_{tun}}{\sum_{i=1}^{T} (b_i + 1) \cdot C_{obj} + E_{logic} \cdot C_{link}}$$

When the physical node scale N is small and the tenant scale T dominates, the logical link terms in the denominator and the physical constants in the numerator are overwhelmed by the scale of growth. To isolate and evaluate the core service instance compression efficiency yielded by distributed anycast gateway deduplication, the above equation can be asymptotically approximated as:

$$CR \approx \frac{\sum_{i=1}^{T} \left(\sum_{j=1}^{b_i} |S_{bd}(i,j)| + |S_{vrf}(i)| \right)}{\sum_{i=1}^{T} (b_i + 1) \cdot C_{obj}}$$

This formula mathematically reveals the fundamental logic behind the resource savings: the numerator represents the summation of redundant service instances forced onto physical devices by distributed gateways due to VM dispersion; the denominator represents the total number of pure, unique logical service objects after deduplication. Regardless of how the tenant scale T varies, since the numerator and denominator remain of the same order, the compression ratio CR stabilizes within a constant range determined by the physical dispersion breadth of the production network's distributed gateways (i.e., the average number of Leaf switches across which each tenant is scattered, $|S_{bd}|$. Conversely, when the physical network scale N (Leaf count) undergoes massive horizontal expansion, the $\frac{1}{2} N^2 \cdot C_{tun}$ term in the numerator dominantly takes over quadratically, causing the CR to skyrocket by orders of magnitude as N increases.

4. Simulation Evaluation and Result Analysis

4.1 Simulation Setup and Key Metrics

To quantitatively compare the evolutionary trends of both models, a large-sample Monte Carlo random simulation evaluation system was implemented (the relevant evaluation algorithm framework and open-source replication scripts are provided in the open-source link at the end of this paper). The baseline simulation parameters were set as follows: the physical network scale ranged within $N \in [10,100]$ (covering scenarios from a single-POD architecture up to the upper limit of ultra-large-scale cloud campus networks); the tenant scale T was scanned from 10 to 400[10]. The number of logical BDs for each tenant followed a uniform random distribution reflecting a typical three-tier architecture, i.e., $b_i \sim U(2,6)$, which simulates the highly disordered scattering of tenant assets across the live network[11]. The relative weight coefficients of different model entities were configured as constants based on the complexity of the underlying framework.

By executing simulation experiments with multi-dimensional independent variables, the operational outcomes of different cross-sections were integrated into a single data chart, from which key physical scale cross-section data were extracted. The final resource control performance is illustrated in Figure 2, with the key cross-sectional data detailed in Table 1.

Table 1 Cross-sectional experimental data under different network scales N

Switch Count N	Full Model S_u (T=50)	Deduplicated Model S_o(T=50)	Full Model S_u(T=300)	Deduplicated Model S_o (T=300)	Compression Ratio CR (T=300)
10	1879	348	9415	2139	4.40
20	4860	328	22122	2127	10.40
30	7195	350	33095	2087	15.86
40	10964	360	45208	2078	21.76
50	15295	346	58747	2113	27.80
60	20286	368	72156	2075	34.77
70	25063	342	84535	2093	40.39
80	30264	347	98584	2078	47.44
90	35991	348	111693	2070	53.96
100	43110	365	129560	2097	61.78

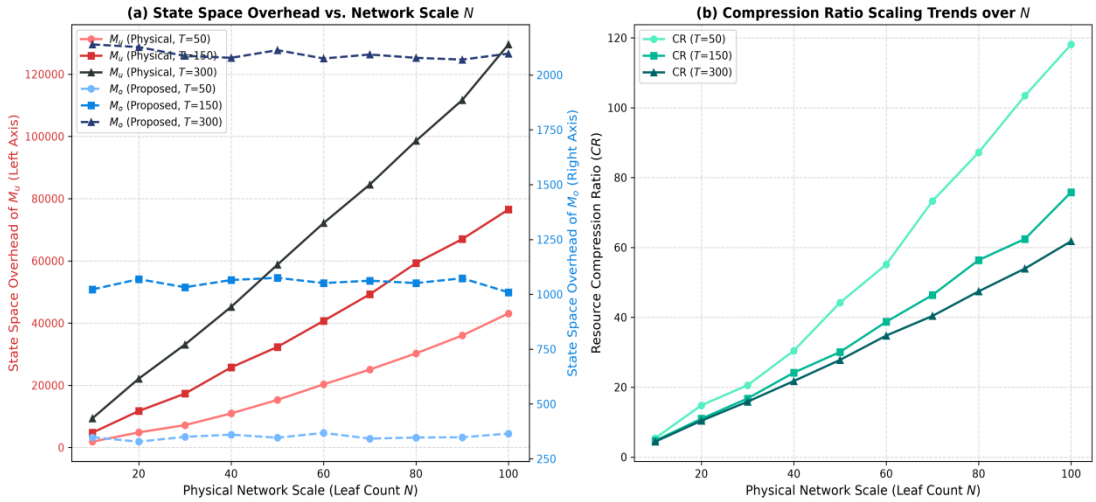


Fig.2 Comparison curves of twin state space resource overheads under multi-dimensional variable evolution

4.2 Interpretation of Experimental Conclusions

By comparing the cross-sectional data in Table 1 with the dynamic curves in Figure 2, two core academic conclusions can be drawn:

①Complete immunity to physical topology expansion ($\mathcal{O}(1)$ property): the $S_o(T = 300)$ column in Table 1, when the underlying network scale N of physical switches surges multi-fold from 10 to an ultra-large cluster of 100, the overhead of the traditional full-scale model S_u grows quadratically due to the joint impact of $\mathcal{O}(N^2)$ physical tunnel states and distributed anycast replica explosions, eventually reaching a state space overhead of 129,560. In sharp contrast, the state space overhead of the proposed Overlay logically deduplicated twin model consistently remains within a narrow corridor around 2,100 (subtly fluctuating from 2,139 to 2,097), which manifests as a perfectly horizontal constant line in Figure 2. This absolute immunity of overhead against physical network horizontal scaling quantitatively validates that the deduplication algorithm successfully eliminates all quadratic and multiplicative terms containing N, thereby empowering the digital twin sandbox with a "dimension-reduction immunity" attribute from a high-level business perspective.

② The larger the physical network scale, the more the deduplication compression gains leap by orders of magnitude: As anticipated by the mathematical model derivation in Chapter 3, with the scaling of the data center's physical topology size N, redundant entries forced onto physical switches by distributed gateways accumulate geometrically. When fixing the tenant scale at T=300, the resource compression ratio (CR) exhibits a steady linear upward trend as the number of physical nodes N increases. In a small-scale private cloud single-POD scenario where N=10, the compression ratio is a modest 4.40-fold; however, when the network horizontally scales to an ultra-large public cloud campus scenario where N=100, the resource compression multiplier scales up to 61.78-fold. This demonstrates that the efficiency of the proposed algorithm does not degrade with the expansion of the network scale. Instead, it exhibits excellent scalability characterized by "the larger the physical network, the more prominent the asset savings," which serves as the efficiency cornerstone for deploying pre-verification sandboxes into large-scale DCs to safeguard the second-level safety closed-loops of LLM-driven IBNs.

5. Conclusion and Future Work

Aiming at the state space explosion problem of digital twins caused by distributed anycast gateways in cloud data centers, this paper proposes a lightweight digital twin modeling method

oriented toward the Overlay logical layer. By establishing a state space mathematical model under equivalent verification constraints, we theoretically prove that the new scheme eliminates all quadratic and multiplicative terms containing the physical node count N by stripping away Underlay physical details and performing high-level logical deduplication, successfully reducing the state space complexity from $\mathcal{O}(N^2 + T \cdot N)$ to $\mathcal{O}(T)$ .Multi-dimensional Monte Carlo random simulation experiments show that while ensuring complete equivalence in connectivity and isolation verification, the proposed deduplicated logical model possesses complete O(1)-level constant immunity against the horizontal expansion of the underlying physical network. In a large-scale network scenario featuring N=100 physical switches and a tenant scale of T=300, this method achieves up to a 61.78-fold compression of the resource space. This approach provides a core theoretical foundation and quantitative experimental support for the lightweight deployment of digital twins in ultra-large-scale data centers.

The ultimate value of the proposed method lies in its role as a millisecond-level pre-deployment safety sandbox validation engine before LLM configurations are pushed to production. In future intent-driven networks, network configuration deployment will undergo a three-stage closed-loop validation process. In the first stage (Intent Translation and Pre-deduction), after a user inputs a natural language business intent, the LLM translates it and generates specific underlying configuration scripts. In the second stage (Lightweight Sandbox Millisecond-level Safety Check), the scripts are fed into the deduplicated Overlay digital twin proposed in this paper. Since this model strips away all redundant physical entities and tunnel encapsulations, the vertices and edges of the graph topology are substantially compressed. Consequently, running routing reachability traversals or access control list (ACL) rule conflict retrievals within the twin environment drops its computational latency from minutes in traditional physical simulations to milliseconds. The sandbox can instantly determine whether the LLM has generated a configuration hallucination and execute either an approval or an interception. In the third stage (Physical Deployment), the logic configurations verified to be hallucination-free are handed over to the authoritative SDN controller, which completes the final distributed 1:N replica instantiation expansion and configuration deployment onto the live physical switches. This closed-loop mechanism, which completely decouples high-level logical validity verification from low-level physical distributed deployment, effectively bridges the native gap between the second-level safety enforcement requirements of large models and the high latency of traditional physical network simulations.

In future research, primary efforts will be dedicated to advancing the intermediate translation and mapping technologies between the digital twin network and the live network, automatically translating the full Underlay network (physical switches, distributed gateways, and full-mesh tunnels) into an Overlay logical digital twin network, thereby completing the automated mapping pipeline between live configurations and the twin model. Through configuration collection, normalization, and deduplication translation from production environments, the lightweight twin can be instantaneously generated based on the actual live topology and incrementally synchronized upon configuration changes; simultaneously, the logic configurations verified in the sandbox will be reversely translated into live physical configurations to interface with the physical deployment phase of the three-stage closed loop, ultimately enabling the millisecond-level pre-deployment check mechanism proposed herein to truly land in large-scale production networks.

Data and Code Availability Statement: The full source code of the Monte Carlo random simulation evaluation system, experimental simulation data, and high-definition academic charts involved in this study are fully open-sourced and hosted on the GitHub repository. Readers and peers can access and execute complete replication via the following link:*https://github.com/qiuqiueyu/Overlay-DigitalTwin-Deduplication*

References

[1] CLEMM A, CIAVAGLIA L, GRANVILLE L Z, et al. Intent-Based Networking - Concepts and Definitions: RFC 9315[S/OL]. IETF, 2022[2026-10-31]. https://www.rfc-editor.org/rfc/rfc9315.html

[2] SAJASSI A, DRAKE J, BITAR N, et al. A Network Virtualization Overlay Solution Using Ethernet VPN (EVPN): RFC 8365[S/OL]. IETF, 2018[2026-10-31]. https://www.rfc-editor.org/rfc/rfc8365.html

[3] LANTZ B, HELLER B, MCKEOWN N. A Network in a Laptop: Rapid Prototyping for Software-Defined Networks[C]//Proceedings of the 9th ACM SIGCOMM Workshop on Hot Topics in Networks (HotNets-IX). New York: ACM, 2010: 1-6.

[4] KREUTZ D, RAMOS F M V, VERISSIMO P, et al. Software-Defined Networking: A Comprehensive Survey[J]. Proceedings of the IEEE, 2015, 103(1): 14-76.

[5] AL-FARES M, LOUKISSAS A, VAHDAT A. A Scalable, Commodity Data Center Network Architecture[C]//Proceedings of the ACM SIGCOMM 2008 Conference on Data Communication (SIGCOMM '08). New York: ACM, 2008: 63-74.

[6] ROY A, ZENG H, BAGGA J, et al. Inside the Social Network's (Datacenter) Network[C]//Proceedings of the 2015 ACM Conference on Special Interest Group on Data Communication (SIGCOMM '15). New York: ACM, 2015: 123-137.

[7] MAHALINGAM M, DUTT D, DUDA K, et al. Virtual eXtensible Local Area Network (VXLAN): A Framework for Overlaying Virtualized Layer 2 Networks over Layer 3 Networks: RFC 7348[S/OL]. IETF, 2014[2026-10-31]. https://www.rfc-editor.org/rfc/rfc7348.html

[8] SAJASSI A, SALAM S, THORIA S, et al. Integrated Routing and Bridging in Ethernet VPN (EVPN): RFC 9135[S/OL]. IETF, 2021[2026-10-31]. https://www.rfc-editor.org/rfc/rfc9135.html

[9] KOPONEN T, AMIDON K, BALLAND P, et al. Network Virtualization in Multi-Tenant Datacenters[C]//Proceedings of the 11th USENIX Symposium on Networked Systems Design and Implementation (NSDI '14). Berkeley: USENIX Association, 2014: 203-216.

[10] SINGH A, ONG J, AGARWAL A, et al. Jupiter Rising: A Decade of Clos Topologies and Centralized Control in Google's Datacenter Network[C]//Proceedings of the 2015 ACM Conference on Special Interest Group on Data Communication (SIGCOMM '15). New York: ACM, 2015: 183-197.

[11] CISCO SYSTEMS. Data Center Multi-Tier Model Design[EB/OL]. Cisco Data Center Infrastructure 2.5 Design Guide[2026-10-31]. https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Data_Center/DC_Infra2_5/DCI_SRND_2_5a_book/DCInfra_2a.html

Author Biographies

Chi Qingyuan, male, Ph.D. student. His research interests focus on Intent-Driven Networks (IBN) powered by Large Language Models, as well as Software-Defined Networking (SDN) and RunOS application. E-mail: chi.q@asvk.cs.msu.ru.

Ruslan L. Smelyanskiy, male, Academician of the Russian Academy of Sciences, Professor, Ph.D. supervisor, and Head of the Chair of Automated Systems Verification and Configuration (ASVK) at the Faculty of Computational Mathematics and Cybernetics, Lomonosov Moscow State University. E-mail: smel@cs.msu.su.

The author highly appreciates the continuous guidance and invaluable support from Prof. Ruslan L. Smelyanskiy, Corresponding Member of the Russian Academy of Sciences and Head of the Automated Systems Verification and Configuration (ASVK) Laboratory at the Faculty of Computational Mathematics and Cybernetics, Lomonosov Moscow State University. His profound insights into network virtualization and digital twins have significantly guided the theoretical formulation and verification framework of this study. The author also thanks all members of the ASVK Laboratory for providing an excellent research environment.)

基于实例去重的云数据中心 Overlay 数字孪生等效建模

Chi Qingyuan[1], Ruslan L. Smeliansky[1]

莫斯科大学计算数学与控制系，莫斯科 119991，俄罗斯 chi.q@asvk.cs.msu.ru

摘要：随着基于大语言模型（LLM）的意图驱动网络（IBN）在云数据中心（DC）的快速发展，数字孪生已成为验证网络意图、防止 LLM 产生配置幻觉的关键沙盒。然而，传统基于物理拓扑的 1:1 全量建模仿真，在面对分布式网关（EVPN-VXLAN）带来的海量物理实例复制时，会导致严重的资源爆炸与仿真延迟，难以满足秒级闭环验证的需求。本文提出了一种面向 Overlay 逻辑层的轻量化数字孪生建模方法。该方法利用 VXLAN 协议对上层业务的透明性，剥离底层物理细节，将现网中分布式散落在多台 Leaf 交换机上的虚拟路由转发（VRF）与桥接域（BD）实例进行“逻辑去重”。本文构建了同等等效约束下的仿真开销数学模型，理论证明了其能将数字孪生网络的状态空间复杂度从 $\mathcal{O}(\mathbf{N^2 + T \cdot N})$ 降至 $\mathcal{O}(\mathbf{T})$，其中N为物理交换机网络规模，T为现网总租户规模。多维蒙特卡洛模拟实验表明：在保障业务连通性与隔离性验证完全等效的前提下，本方法在包含 100 台 Leaf 交换机及 300 个租户的超大规模网络场景下实现了高达 61.78 倍的资源空间压缩，且状态空间对物理节点数变动具备完全的$\mathcal{O}(\mathbf{1})$级常数免疫特性。该轻量化基座显著提升了数字孪生的验证效率，为未来基于大模型驱动的意图网络迈向秒级闭环、防幻觉的自主运行阶段奠定了扎实的工业应用基座。

关键词： 数字孪生；意图驱动网络（IBN）；Overlay 抽象；逻辑实例去重

Equivalent Modeling of Cloud Data Center Overlay Digital Twin Based on Instance Deduplication

Abstract: With the rapid evolution of Intent-Driven Networks (IBN) powered by Large Language Models (LLMs) in cloud data centers (DCs), digital twins have emerged as a critical sandbox to verify network intents and prevent LLM hallucinations. However, traditional 1:1 full-scale simulation based on Underlay physical topologies suffers from severe resource explosion and verification latency due to the massive multiplication of physical instances introduced by distributed anycast gateways (EVPN-VXLAN). To achieve second-level closed-loop verification, this paper proposes a lightweight digital twin modeling method oriented toward the Overlay logical layer. Capitalizing on the transparency of VXLAN protocols to upper-layer services, this approach strips away underlying physical details and performs "logical deduplication" on Virtual Routing and Forwarding (VRF) and Bridge Domain (BD) instances scatteredly deployed across multiple physical Leaf switches. We construct a mathematical model for simulation overhead under equivalent verification constraints, theoretically demonstrating that the state space complexity is reduced from $O(N^2+T\cdot N)$ to $O(T)$, where N denotes the physical network scale and T denotes the total tenant scale. Multi-dimensional Monte Carlo simulations show that while ensuring complete equivalence in connectivity and isolation verification, the proposed

method achieves up to a 61.78-fold resource space compression in a large-scale network scenario featuring 100 Leaf switches and 300 tenants, rendering the twin state space fully immune to physical node expansion with an $O(1)$ constant property. This lightweight infrastructure significantly improves the verification efficiency of digital twins, providing an ultra-fast and low-overhead anti-hallucination verification foundation for future LLM-driven IBN toward second-level autonomous closed-loop operation.

Key words: Digital Twin; Intent-Driven Network (IBN); Cloud Data Center; Overlay Abstraction; Logical Instance Deduplication; Large Language Models

1. 引言

近年来，意图驱动网络（IBN）与大语言模型（LLM）的深度融合成为下一代网络智能运维（NetOps）的核心方向[1]。LLM 凭借强大的自然语言理解能力，能将抽象业务意图直接转化为网络配置脚本。然而，生成式大模型固有的“幻觉”问题，使其生成的配置指令极易包含路由冲突、隔离失效等高危风险。因此，配置正式下发前，必须在数字孪生仿真环境中进行前置合规验证。

在现代云数据中心场景下，网络普遍采用基于分布式网关架构的 Spike-Leaf 拓扑（如 EVPN-VXLAN）[2]。为了支持虚拟机（VM）在任意物理机上的大二层漫游，同一租户的虚拟私有云（VPC）实例、三层路由转发域（VRF）以及二层桥接域（BD）会高度散落并实例化在多台物理 Leaf 交换机上。在传统 1:1 的 Underlay 物理全量数字孪生建模中，为了验证配置的正确性，必须为每台物理设备启动对应的网络操作系统镜像或进程，并完全仿真底层全网状（Full-Mesh）的 VXLAN 隧道[3]。

这种传统的全量模拟面临着严重的“物理副本爆炸”问题：举例而言，假设某云数据中心内一个典型租户（VPC）设计有 1 个逻辑 VRF 和 2 个逻辑 BD 子网。BD1 下挂载 4 个 VM，BD2 下挂载 3 个 VM。由于云计算调度的随机性，这 7 个 VM 恰好散落在 7 台不同的物理 Leaf 交换机上。若采用 1:1 物理全量模拟，系统必须在 7 台虚拟交换机实例上各自运行 1 份完全相同的 VRF 进程、BD 实例以及 IRB 三层接口配置。当全网包含 T 个租户，且每个租户的 VM 高度零散分布时，孪生沙盒内的物理配置实例数量将随租户数与设备数的乘积呈级数爆炸。这导致数字孪生的构建需要耗费海量的计算硬件资源，验证周期被拉长至分钟级，彻底失去了护航大模型 IBN 进行秒级意图闭环的价值。

基于此，本文提出一种面向 Overlay 逻辑层的轻量化数字孪生建模方法。本文的核心贡献在于：

① 提出了剥离底层 Underlay 物理依赖、基于逻辑实例去重的云数据中心轻量化 Overlay 数字孪生架构，打破了仿真开销与物理交换机规模的耦合。

② 建立了同等业务等效约束下的状态空间开销数学模型，严密证明了新方法能将复杂度从 $\mathcal{O}(N^2 + T \cdot N)$ 降至 $\mathcal{O}(T)$。

③ 通过多维度的蒙特卡洛随机仿真实验，定量证实了该方法在大规模网络环境下能够实现百倍以上的资源状态空间压缩，且对物理拓扑扩展具有完全的免疫性。

2. 架构映射与业务等效性

2.1 轻量化 Overlay 映射架构与去重思路

本文提出的轻量化建模原则是“抽象底层、去重高层、业务等效”。该方法打破了传统“先构建物理设备，再下发业务配置”的逆向建模思路，直接以高层业务实体 VPC、VRF、BD 为基础构建逻辑孪生体。具体映射原则如下：将租户的虚拟私有云（VPC）抽象为全网唯一的、去重后的三层路由隔离域逻辑节点（逻辑 VRF）。将 VPC 内的子网抽象为去重后的二层广播域（逻辑 BD）逻辑节点。虚拟机（VM）作为终端节点，解耦物理宿主机标签，直接挂载在对应的逻辑 BD 节点下。

在轻量化 Overlay 孪生体中，彻底抹去物理 Leaf 交换机实体以及它们之间两两互联的全网状物理 VXLAN 隧道。现网中各 VRF/BD 究竟散落在哪些具体物理 Leaf 上的映射元数据，交由 SDN 控制器进行内存级表项维护，不在数字孪生体内作为独立运行实体展开[4]。

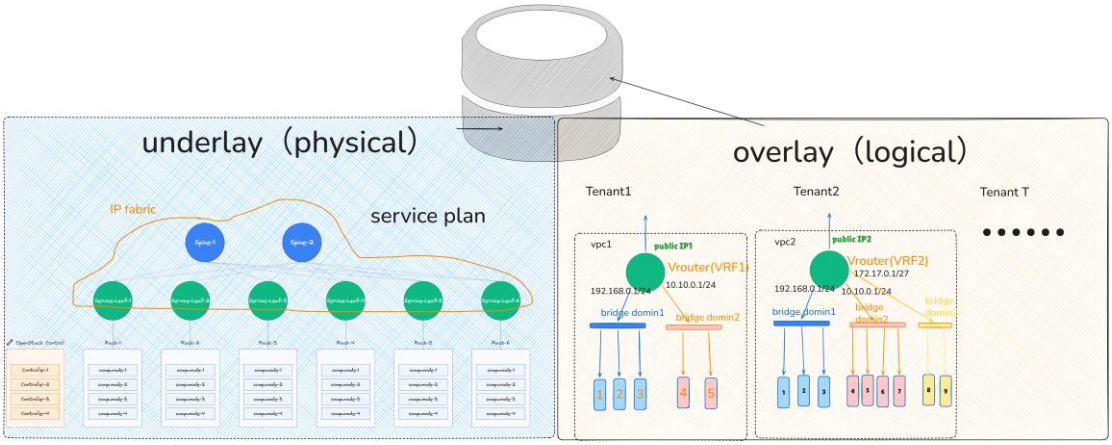


图 1 云数据中心 Underlay 1:1 物理全量建模与 Overlay 逻辑去重建模架构对比及思路示意图

2.2 业务等效严格定义与简化合理性

为了确保轻量化模型在丢弃底层细节后仍能精准拦截大模型的配置幻觉，本文给出“业务等效”的数学严格定义。

设大模型（LLM）生成的待验证网络安全意图策略任务集为 $\boxtimes$（包括检查租户间 VPC 强隔离性、子网内部连通性、跨网段路由可达性、ACL 访问控制策略合规性等）。设传统全量物理数字孪生模型为 M_u，其对应的意图判定函数为 $f_u(r), r \in \boxtimes$；设本文提出的去重逻辑数字孪生模型为 M_o，其判定函数为 $f_o(r)$。当且仅当两模型对任意任务的判定结果完全一致时，称两模型业务等效：

$$\mathrm{Equiv}(M_u, M_o) \Leftrightarrow \forall r \in \boxtimes : f_u(r) = f_o(r)$$

本模型的核心前置假设是“忽略 Underlay 链路拥塞与底层封装细节”。这一假设基于现代云数据中心的工业界运行现状：数据中心普遍采用高冗余的两层或三层 CLOS 拓扑，并运行 ECMP 进行全网流量均衡负载分摊[5]。工业界现网测量数据表明，DC 底层链路利

用率总体处于低负载安全区间，边缘链路平均利用率通常低于 10%，汇聚层链路利用率中位数约 10%~20%[6]。由于 VXLAN 封装本身对上层业务数据流完全透明，底层路由的扰动和物理封装并不改变上层业务配置的逻辑对错（如错配 IP、漏配 ACL 等）[7]。因此，剥离底层细节并不会引发上层业务连通性与隔离性判定准确率的跌落。

3. 仿真开销数学模型

为了从理论上严格证明资源节省的根本原因，本节建立两种孪生模型的渐近复杂度状态空间模型。假设网络中共有 N 个物理 Leaf（VTEP）节点，当前全网租户（VPC）总数为 T。对于第 i 个租户（$1 \le i \le T$），其业务规划拥有的逻辑 BD 数量为 b_i。

3.1 传统 Underlay1:1 全量模型开销 (M_u)

在全量物理仿真中，数字孪生系统必须为每个租户的每个网关实例在其物理上实际散落的 Leaf 交换机上分别创建副本[8]。在最坏情况下，租户业务充分散落漫游，设第 i个租户的第 j个 BD 实际分布的 Leaf 交换机集合为$S_{bd}(i,j)$，其对应的三层 VRF 实际分布的 Leaf 交换机集合为 $S_{vrf}(i)$。则全量模型 M_u 在运行时的总状态空间资源开销S_u为：

$$S_u = \sum_{i=1}^{T} \left(\sum_{j=1}^{b_i} |S_{bd}(i,j)| \right) + \sum_{i=1}^{T} |S_{vrf}(i)| + N \cdot C_{leaf} + \frac{N(N-1)}{2} \cdot C_{tun}$$

式中，$|\cdot|$ 代表集合的基数（即该业务实例散落的物理 Leaf 交换机绝对数量）；C_{leaf} 为单台物理交换机本身 NOS 操作系统的基础常数开销；$\frac{N(N-1)}{2}$ 为 N 个物理节点两两互联所产生的全网状（Full-Mesh）VXLAN 隧道总数，而 C_{tun} 则为每条物理隧道在孪生数据面维持所需的动态协议状态常数开销。

根据分布式 Anycast 网关的现网特征，随着网络物理规模 N 与租户规模 T 同步扩大，网关在物理层面的分布项 $|S_{bd}|$ 将趋近于 N，导致公式前两项的物理实例配置副本数以 $\mathcal{O}(T \cdot N)$ 的复合规模膨胀；同时，最后一项物理隧道开销由于 Full-Mesh 特征，呈现刚性的 $\mathcal{O}(N^2)$ 平方级增长。因此，全量模型 M_u 的整体状态空间复杂度上界为：

$$S_u = \mathcal{O}(N^2 + T \cdot N)$$

3.2 本文 Overlay 逻辑去重模型开销 (M_o)

在 Overlay 孪生模型 M_o 中，由于从底层解耦了全部物理节点与全网状隧道，且对任意物理交换机上的相同配置进行了全网业务去重，不论该 BD 或 VRF 在现网分布式部署得有多广，在孪生沙盒中均仅唯一实例化 1 次。其状态空间总开销 S_o 表达为：

$$S_o = \sum_{i=1}^{T} (b_i + 1) \cdot C_{obj} + E_{logic} \cdot C_{link}$$

其中，C_{obj} 为单个纯逻辑业务对象（VRF 或 BD）在去重基座上的单位常数开销；E_{logic} 为租户内部虚拟机（VM）到逻辑 BD 之间的基础拓扑逻辑链路总数；C_{link} 为单条逻辑链路的维护开销常数。

由于在实际数据中心中，单个租户定义的子网数 b_i 和 VM 数量是由用户自身业务架构决定的，属于不随物理设备规模 N 变动的有限小常数（通常为 $2 \sim 6$ 之间的常数）[9]。因此，可以将单个租户的平均逻辑开销合并为常数 K，公式简化为：

$$S_o = \sum_{i=1}^{T} K = K \cdot T$$

由此理论证明，去重逻辑模型 M_o 的空间复杂度呈现严格的租户规模线性正相关，且与物理交换机数量 N 彻底解耦：

$$S_o = \mathcal{O}(T)$$

3.3 理论压缩比推导

为了直观评估本轻量化方法对孪生体状态空间的压缩效率，本文定义理论压缩比（Compression Ratio, CR）为传统全量模型开销 S_u 与本文 Overlay 逻辑去重模型开销 S_o 的比值：

$$CR = \frac{S_u}{S_o} = \frac{\sum_{i=1}^{T}\left(\sum_{j=1}^{b_i} |S_{bd}(i,j)|\right) + \sum_{i=1}^{T} |S_{vrf}(i)| + N \cdot C_{leaf} + \frac{N(N-1)}{2} \cdot C_{tun}}{\sum_{i=1}^{T}(b_i + 1) \cdot C_{obj} + E_{logic} \cdot C_{link}}$$

当网络物理节点规模 N 较小且租户规模 T 占据主导时，分母中的逻辑链路项与分子中的物理常数项由于规模增长而被淹没。此时，为了直观评估因分布式 Anycast 网关去重带来的核心业务实例压缩效率，上式可近似约简为：

$$CR \approx \frac{\sum_{i=1}^{T}\left(\sum_{j=1}^{b_i} |S_{bd}(i,j)| + |S_{vrf}(i)|\right)}{\sum_{i=1}^{T}(b_i + 1) \cdot C_{obj}}$$

该公式从数学上揭示了资源节省的根本逻辑：分子代表在分布式网关下，各租户的业务实例因为虚机充分散落而被物理设备强制复制的总和。分母代表去重后，全网唯一的、纯粹的业务逻辑对象总和。无论租户数 T 如何变动，该压缩比 CR 将由于上下分子分母同阶，稳定由现网分布式网关的物理散落广度（即平均每租户散落的 Leaf 数量 $|S_{bd}|$）决定的常数区间。而当网络物理规模 N（Leaf 数量）大幅横向扩展时，分子中的 $\frac{1}{2}N^2 \cdot C_{tun}$ 项将呈平方级占据主导，此时 CR 将随 N 的扩大呈现数量级跃升。

4. 仿真评估与结果分析

4.1 仿真设置与关键指标

为了定量对比两者的演进趋势，本文编写了基于大样本的蒙特卡洛随机模拟评估系统（相关评估算法框架及开源复现脚本详见文末开源链接）。设定基准仿真参数：网络物理规模范围设为 $N \in [10,100]$（覆盖单 POD 架构到超大规模云园区网络上限）；租户规模 T 扫描范围设为 10 ~ 400[10]。每个租户的逻辑 BD 数量设为满足典型三层架构的随机分布

$b_i \sim U(2,6)$，模拟租户资产高度无序散落现网[11]。各模型实体的相对权重系数根据系统底座复杂度设为常数。

通过运行多维自变量变动的仿真实验，我们在同一张数据图表中综合集成了两种自变量变动截面的运行结果，并提取了关键物理规模截面数据，其展现出的最终资源控制效能如图 2 所示。

表 1 不同网络规模（N）下的实验数据

交换机数量 N	全量模型 S_u (T=50)	去重模型 S_o(T=50)	全量模型 S_u(T=300)	去重模型 S_o (T=300)	压缩比CR (T=300)
10	1879	348	9415	2139	4.40
20	4860	328	22122	2127	10.40
30	7195	350	33095	2087	15.86
40	10964	360	45208	2078	21.76
50	15295	346	58747	2113	27.80
60	20286	368	72156	2075	34.77
70	25063	342	84535	2093	40.39
80	30264	347	98584	2078	47.44
90	35991	348	111693	2070	53.96
100	43110	365	129560	2097	61.78

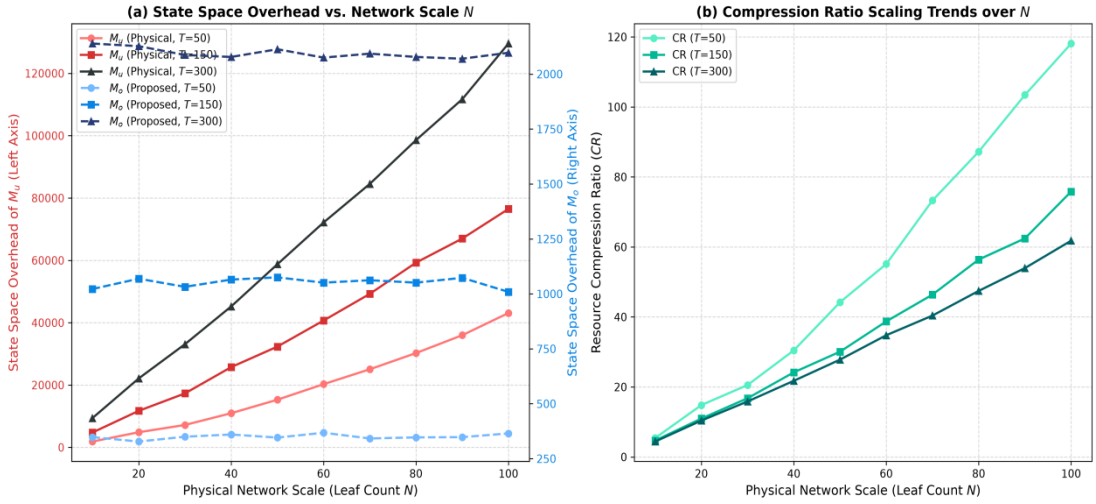


图 2 多维变量演进下的孪生体状态空间资源开销对比曲线

4.2 实验结论解读

通过多维蒙特卡洛仿真的横截面数据（如表 1 所示）与图 2 的动态曲线对比，可以得出以下两点核心学术结论：

①对物理拓扑扩展具有完全的免疫性（$\mathcal{O}(1)$ 属性）：
观察表 1 中的 $S_o(T=300)$ 一列，当底层物理交换机网络规模N从 10 台成倍激增到 100 台超大集群时，传统全量模型的开销S_u受到 $\mathcal{O}(N^2)$ 物理隧道状态与分布式 Anycast 副本爆炸的双重影响，呈现二次方级的快速增长，最终增至 129,560 的状态空间开销。与之形成鲜明对比的是，本文提出的 Overlay 逻辑去重孪生模型开销却始终平稳地保持在 2,100 左右的窄幅区间内（从 2,139 微幅波动至 2,097），在图 2 中表现为一条绝对水平的常数直线。这种开销对物理网络横向扩容的完全免疫性，从定量层面有力验证了去重算法成功消除了所有包含 N 的平方项与乘数项，使数字孪生沙盒具备了高阶业务视角的“降维免疫”属性。

②网络物理规模越大，去重压缩收益呈数量级跃升：
正如第三章数学模型推导的预期，随着数据中心物理拓扑规模 N 的扩大，分布式网关在物理交换机上强制复制的冗余表项将呈几何级数堆叠。当固定租户规模 $T=300$ 时，随着网络物理节点 N 的增加，资源的压缩比（CR）呈现稳定的线性上升趋势。在 $N=10$ 的私有云单 POD 小规模场景下，压缩比仅为 4.40 倍；而当网络横向扩展至$N=100$的超大公有云园区场景时，资源压缩倍数提升至 61.78 倍。 这表明本算法的高效性不会因网络规模的扩大而衰减，反而表现出“物理网络越大、节省资产越显著”的良好可扩展性，这正是让前置仿真沙盒能够真正落地于大规模 DC、护航大模型 IBN 秒级安全闭环的效率基石。

5. 结论与未来展望

针对云数据中心分布式 Anycast 网关导致的数字孪生状态空间爆炸问题，本文提出了一种面向 Overlay 逻辑层的轻量化数字孪生建模方法。通过建立同等业务等效约束下的状态空间数学模型，从理论上证明了新方案通过剥离 Underlay 物理细节与高层逻辑去重，成功消除了所有包含物理节点数 N 的二次方项与乘数项，将状态空间复杂度从 $\mathcal{O}(N^2+T\cdot N)$ 压缩至 $\mathcal{O}(T)$。多维蒙特卡洛随机仿真实验表明，在保障业务连通性与隔离性验证完全等效的前提下，本文提出的去重逻辑模型对底层物理网络的横向扩容具备完全的 $\mathcal{O}(1)$ 级常数免疫特性。在大规模网络场景下，当物理设备节点数 $N=100$ 且租户规模 $T=300$ 时，该方法成功实现了高达 61.78 倍的资源空间压缩。本方案为超大规模数据中心数字孪生的轻量化落地提供了核心理论依据与定量的实验支撑。

本文方法的最终价值在于作为大语言模型（LLM）配置下发前的毫秒级前置安检沙盒验证底座。在未来的意图驱动网络中，网络配置的下发将经历三个阶段的闭环验证。第一阶段进行意图翻译与前置推演，用户输入自然语言业务意图后，由 LLM 将其翻译并生成底层的具体配置脚本。第二阶段进行轻量沙盒毫秒级安检，将脚本输入本文的去重 Overlay 数字孪生体中，由于本模型剥离了所有冗余的物理实体与隧道封装，使图拓扑的顶点和边被大幅压缩，此时在孪生体内运行路由可达性遍历或访问控制列表规则冲突检索，其计算耗时将从传统物理仿真的分钟级大幅下降至毫秒级，沙盒能够瞬间判定 LLM 是否产生了配置幻觉并执行放行或拦截。第三阶段进行物理下发，验证无幻觉后的逻辑配置将交由权威

SDN 控制器，由控制器完成最终面向现网物理交换机的分布式 1:N 副本实例化展开与配置下发。这种将高层逻辑合理性验证与底层物理分布式下发彻底解耦的闭环机制，有效填补了大模型秒级安全生效需求与传统物理网络仿真高延迟之间的天然鸿沟。

在未来的研究中，将重点推进数字孪生网络与现网之间的中间映射翻译技术实现，把 Underlay 全网（物理交换机、分布式网关及全网状隧道）自动翻译映射为 Overlay 逻辑数字孪生网络，打通现网配置与孪生模型之间的自动化映射链路。该技术通过现网配置的采集、归一化与去重翻译，使轻量化孪生体能够基于现网真实拓扑即时生成并随配置变更增量同步；同时将沙盒验证后的逻辑配置反向翻译为现网物理配置，与前文三阶段闭环中的物理下发环节相衔接，从而让本文所提的毫秒级前置安检机制真正落地于大规模生产网络。

数据与代码可用性声明：本文研究所涉及的蒙特卡洛随机模拟评估系统、仿真实验数据以及高清学术图表的完整源代码已完全开源，托管于 GitHub 开源平台。读者与同行可通过访问以下路径获取并执行完全复现：*https://github.com/qiuqiueyu/Overlay-DigitalTwin-Deduplication* 。

参考文献

[1] CLEMM A, CIAVAGLIA L, GRANVILLE L Z, et al. Intent-Based Networking - Concepts and Definitions: RFC 9315[S/OL]. IETF, 2022[2026-10-31]. https://www.rfc-editor.org/rfc/rfc9315.html

[2] SAJASSI A, DRAKE J, BITAR N, et al. A Network Virtualization Overlay Solution Using Ethernet VPN (EVPN): RFC 8365[S/OL]. IETF, 2018[2026-10-31]. https://www.rfc-editor.org/rfc/rfc8365.html

[3] LANTZ B, HELLER B, MCKEOWN N. A Network in a Laptop: Rapid Prototyping for Software-Defined Networks[C]//Proceedings of the 9th ACM SIGCOMM Workshop on Hot Topics in Networks (HotNets-IX). New York: ACM, 2010: 1-6.

[4] KREUTZ D, RAMOS F M V, VERISSIMO P, et al. Software-Defined Networking: A Comprehensive Survey[J]. Proceedings of the IEEE, 2015, 103(1): 14-76.

[5] AL-FARES M, LOUKISSAS A, VAHDAT A. A Scalable, Commodity Data Center Network Architecture[C]//Proceedings of the ACM SIGCOMM 2008 Conference on Data Communication (SIGCOMM '08). New York: ACM, 2008: 63-74.

[6] ROY A, ZENG H, BAGGA J, et al. Inside the Social Network's (Datacenter) Network[C]//Proceedings of the 2015 ACM Conference on Special Interest Group on Data Communication (SIGCOMM '15). New York: ACM, 2015: 123-137.

[7] MAHALINGAM M, DUTT D, DUDA K, et al. Virtual eXtensible Local Area Network (VXLAN): A Framework for Overlaying Virtualized Layer 2 Networks over Layer 3 Networks: RFC 7348[S/OL]. IETF, 2014[2026-10-31]. https://www.rfc-editor.org/rfc/rfc7348.html

[8] SAJASSI A, SALAM S, THORIA S, et al. Integrated Routing and Bridging in Ethernet VPN (EVPN): RFC 9135[S/OL]. IETF, 2021[2026-10-31]. https://www.rfc-editor.org/rfc/rfc9135.html

[9] KOPONEN T, AMIDON K, BALLAND P, et al. Network Virtualization in Multi-Tenant Datacenters[C]//Proceedings of the 11th USENIX Symposium on Networked Systems Design and Implementation (NSDI '14). Berkeley: USENIX Association, 2014: 203-216.

[10] SINGH A, ONG J, AGARWAL A, et al. Jupiter Rising: A Decade of Clos Topologies and Centralized Control in Google's Datacenter Network[C]//Proceedings of the 2015 ACM Conference on Special Interest Group on Data Communication (SIGCOMM '15). New York: ACM, 2015: 183-197.

[11] CISCO SYSTEMS. Data Center Multi-Tier Model Design[EB/OL]. Cisco Data Center Infrastructure 2.5 Design Guide[2026-10-31]. https://www.cisco.com/c/en/us/td/docs/solutions/Enterprise/Data_Center/DC_Infra2_5/DCI_SRND_2_5a_book/DCInfra_2a.html

作者简介

• 迟庆元，男，博士研究生，研究方向为基于大语言模型（LLM）与软件定义网络（SDN，基于 RunOS 控制器）的意图驱动网络（IBN）。电子信箱：chi.q@asvk.cs.msu.ru。

• 鲁斯兰·里昂尼多维奇·斯梅良斯基（通信作者），男，俄罗斯科学院院士，教授，博士生导师，莫斯科大学计算数学与控制系计算系统自动化验证与配置（ASVK）教研室主任。电子信箱：smel@cs.msu.su。